Chapter 13

Operational Amplifiers (Op-Amps): Theory & Characteristics

13.1 Introduction

In the previous chapters, we analyzed amplifiers built from discrete components like BJTs and FETs. While effective, discrete amplifiers suffer from variations in gain due to temperature changes and component tolerances (β variation). Designing a stable, high-gain amplifier with discrete parts is complex.

The **Operational Amplifier (Op-Amp)** solves this. It is a high-performance linear integrated circuit (IC) containing dozens of transistors, resistors, and capacitors on a single silicon chip. Originally designed for analog computers to perform mathematical operations (addition, subtraction, integration, differentiation—hence the name ”Operational”), the Op-Amp is now the fundamental building block of modern analog electronics. It acts as a near-perfect Voltage Controlled Voltage Source (VCVS).

Learning Objectives

By the end of this chapter, the student should be able to:

- Describe the internal architecture (block diagram) of an Op-Amp.
- Distinguish between Ideal and Practical (LM741) characteristics.
- Define and calculate CMRR, Slew Rate, and Input Offset Voltage.
- Analyze the concept of the Virtual Ground.
- Interpret the Open-Loop Frequency Response (Bode Plot).

13.2 The Op-Amp Symbol and Terminals

An Op-Amp typically has five active terminals:

1. **Inverting Input (-):** Signals applied here appear inverted (180° phase shift) at the output.
2. **Non-Inverting Input (+):** Signals applied here appear in-phase (0°) at the output.
3. **Output (V_{out}):** The amplified signal.
4. **Positive Power Supply ($+V_{CC}$):** Usually $+5$V to $+15$V.
5. **Negative Power Supply ($-V_{EE}$):** Usually -5V to -15V.

Note that the Op-Amp amplifies the **difference** between the two inputs:

$$V_{out} = A_{OL}(V_+ - V_-) \tag{13.1}$$

Where A_{OL} is the Open Loop Voltage Gain.

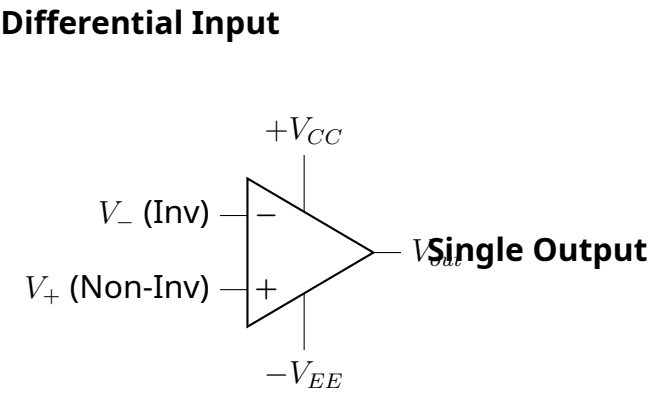


Figure 13.1: Standard Circuit Symbol of an Operational Amplifier.

13.3 13.3 Internal Architecture

What is inside the black chip? A typical Op-Amp (like the 741) consists of three main stages:

13.3.1 Stage 1: The Differential Amplifier

This is the input stage. It consists of a **Dual-Input, Balanced-Output Differential Amplifier** (usually a matched pair of BJTs or FETs).

- **Function:** It provides high input impedance and amplifies the difference between the two input signals while rejecting noise common to both (High CMRR).
- **Gain:** Moderate voltage gain.

13.3.2 Stage 2: Voltage Amplification Stage (VAS)

Also called the Level Shifter or High-Gain Stage.

- **Function:** This stage provides the bulk of the voltage gain ($> 200,000$). It usually consists of high-gain Darlington pairs.
- **Level Shifting:** It also shifts the DC level of the signal so that the output can swing positive and negative around 0V.

13.3.3 Stage 3: Output Stage (Push-Pull)

This is usually a Class B or Class AB Complementary Symmetry Push-Pull amplifier (Emitter Follower).

- **Function:** Provides low output impedance and sufficient current drive capability to drive loads (like LEDs or other circuits) without the voltage dropping.

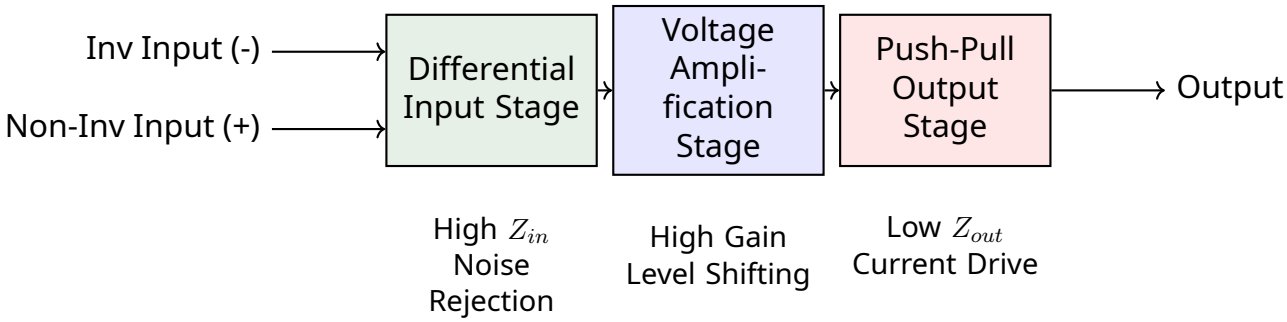


Figure 13.2: Internal Block Diagram of a typical Op-Amp.

13.4 13.4 The 741 Op-Amp

The μA741 (introduced by Fairchild in 1968) is the most famous Op-Amp in history. While modern Op-Amps are faster and more precise, the 741 remains the standard for teaching. It comes in an 8-pin Dual In-line Package (DIP).

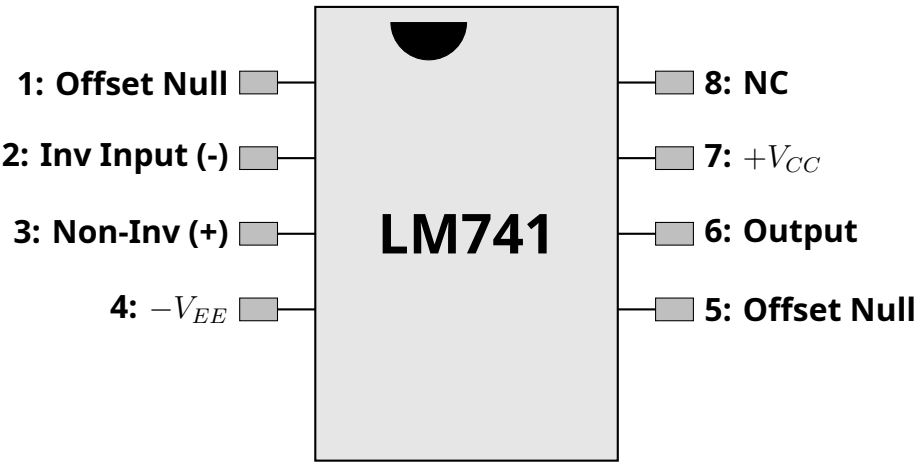


Figure 13.3: Pinout of the LM741 Op-Amp (Top View).

13.5 Ideal vs. Practical Characteristics

To design circuits, we often assume the Op-Amp is "Ideal". However, in high-precision agricultural instrumentation, the deviations of the "Practical" Op-Amp must be considered.

Table 13.1: Comparison of Ideal and Practical (741) Op-Amps

Parameter	Ideal Value	Practical (LM741)
Open Loop Gain (A_{OL})	Infinite (∞)	Very High (2×10^5)
Input Impedance (Z_{in})	Infinite (∞)	High ($2M\Omega$)
Output Impedance (Z_{out})	Zero (0Ω)	Low (75Ω)
Bandwidth (BW)	Infinite (∞)	Limited (1 MHz)
Offset Voltage (V_{os})	Zero (0 V)	Small ($2-6$ mV)
CMRR	Infinite (∞)	90 dB
Slew Rate	Infinite (∞)	0.5 V/μs

13.6 Detailed Electrical Parameters

13.6.1 Input Offset Voltage (V_{os})

Ideally, if both inputs are connected to ground ($V_+ = V_- = 0$), the output should be zero. In reality, due to slight mismatches in the manufacturing of the internal transistors, a small DC error voltage appears at the output.

- V_{os} is the differential input voltage required to force the output to zero.
- Effect: In precision DC amplifiers (like a thermocouple thermometer), this offset is amplified and causes significant measurement errors.
- Correction: Most Op-Amps (like the 741) have "Offset Null" pins (1 and 5) to trim this error to zero using a potentiometer.

13.6.2 Input Bias Current (I_B)

Ideally, input impedance is infinite, so no current flows into the inputs. Practically, the input transistors (BJTs) require a small Base current to operate.

$$I_B = \frac{I_{B+} + I_{B-}}{2}$$

This current (approx 80 nA for 741) flows through external resistors, creating unwanted voltage drops that look like input signals.

13.6.3 Common Mode Rejection Ratio (CMRR)

This defines the Op-Amp's ability to reject noise. Noise (like 50Hz hum from mains wiring) usually appears on both inputs with equal amplitude.

- Differential Gain (A_d): Gain for the difference signal ($V_+ - V_-$).
- Common Mode Gain (A_{cm}): Gain for the common signal (V_{cm}). ideally zero.

$$\text{CMRR} = \frac{A_d}{A_{cm}} \tag{13.2}$$

Usually expressed in decibels:

$$\text{CMRR}_{dB} = 20 \log_{10} \left(\frac{A_d}{A_{cm}} \right) \tag{13.3}$$

A high CMRR (e.g., 90dB) means the amplifier ignores noise and amplifies only the sensor signal.

13.7 AC Performance: Slew Rate & Bandwidth

13.7.1 Slew Rate (SR)

The Slew Rate is the maximum rate at which the output voltage can change. It is limited by the internal capacitors charging speed.

$$\text{SR} = \left.\frac{dV_{out}}{dt}\right|_{max} \quad [\text{Volts}/\mu s] \tag{13.4}$$

For a 741, SR is $0.5V/\mu s$. If you input a high-frequency square wave, the output will look like a trapezoid or triangle because the output cannot rise fast enough.

Example 1: Slew Rate Limitation

An Op-Amp with $SR = 0.5V/\mu s$ is asked to amplify a square wave from 0V to 10V. How long does the output take to rise?

Solution:

$$dt = \frac{dV}{SR} = \frac{10\ \text{V}}{0.5\ \text{V}/\mu s} = 20\mu s$$

Instead of a vertical edge, the pulse will have a rising slope lasting $20\mu s$.

13.7.2 Gain-Bandwidth Product (GBP)

Op-Amps have an internal "Compensation Capacitor" to prevent oscillation. This causes the Open Loop Gain (A_{OL}) to drop as frequency increases. The product of Gain and Bandwidth is constant.

$$\text{GBP} = A_v \times f = \text{Constant} \tag{13.5}$$

For a 741, GBP is 1 MHz.

- If you need a Gain of 1, the Bandwidth is 1 MHz.
- If you need a Gain of 100, the Bandwidth drops to 10 kHz ($1\ \text{MHz}/100$).

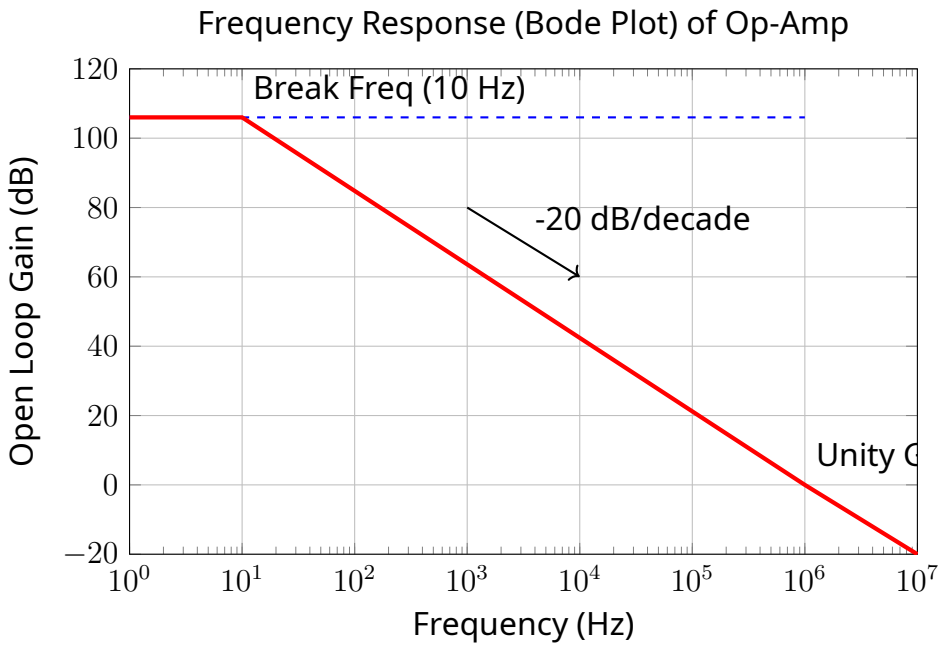


Figure 13.4: Open Loop Frequency Response. Gain decreases as frequency increases.

13.8 The Concept of Negative Feedback

An ideal Op-Amp has infinite gain. If we use it in **Open Loop** (no connection from output to input), the slightest noise at the input will drive the output to saturation (+V_{CC} or -V_{EE}). This makes it useless as a linear amplifier.

To tame this infinite gain, we use **Negative Feedback**: feeding a portion of the output signal back to the **Inverting (-)** input.

13.8.1 Benefits of Negative Feedback

Although negative feedback reduces the overall gain, it provides:

1. **Stability:** Gain is determined by external resistors (R_f, R_{in}), not the transistor properties.
2. **Bandwidth Extension:** Lower gain means higher bandwidth (GBP is constant).
3. **Linearity:** Reduces distortion.
4. **Input/Output Impedance Control:** Can increase Z_{in} and decrease Z_{out}.

13.9 The Virtual Ground Concept

When analyzing Op-Amp circuits with negative feedback, we use two fundamental rules (The Golden Rules):

The Golden Rules

1. **Current Rule:** No current flows into the input terminals ($I_+ = I_- = 0$). (Due to infinite Z_{in}).
2. **Voltage Rule:** The Op-Amp will adjust its output to make the voltage difference between inputs zero ($V_+ = V_-$).

Virtual Ground Explanation: In an Inverting Amplifier, the Non-Inverting input (+) is connected to Ground (0V). According to the Voltage Rule, the Op-Amp tries to keep the Inverting input (-) at the same potential.

$$V_+ = 0\,\text{V} \implies V_- = 0\,\text{V}$$

However, the Inverting input is NOT physically connected to the ground. It is held at 0V by the feedback mechanism. This node is effectively a **Virtual Ground**. This concept greatly simplifies circuit analysis.

13.10 Chapter Summary

1. The Op-Amp is a high-gain, differential voltage amplifier.
2. **Block Diagram:** Input Diff Amp $\rightarrow$ Voltage Amp $\rightarrow$ Output Stage.
3. **Ideal Characteristics:** Infinite Gain, Infinite Input Z, Zero Output Z.
4. **Practical Limitations:** Offset Voltage, Bias Currents, Slew Rate limits, and finite Bandwidth.
5. **Negative Feedback** is essential for stable, linear operation.

6. **Virtual Ground:** A key analysis tool where $V_- \approx V_+$ due to high gain.

Review Questions

Conceptual Questions

1. Why is the input stage of an Op-Amp always a Differential Amplifier?
2. Explain the term ”Virtual Ground”. Does current flow into a virtual ground?
3. Define ”Slew Rate”. How does it affect the output if we try to amplify a very high-frequency square wave?
4. Why does the Open-Loop gain of a 741 Op-Amp decrease as frequency increases?
5. Differentiate between Differential Mode Gain and Common Mode Gain. Which one do we want to be higher?

13.10.1 Advanced Calculation Questions

6. **Slew Rate Calculation:** An Op-Amp with a Slew Rate of $0.5V/\mu s$ is used to amplify a sinusoidal signal $V_{in} = 2\sin(2\pi ft)$ with a gain of 10. Calculate the maximum frequency (f_{max}) that can be amplified without distortion. *(Hint: Max slope of sine wave $V_p \sin(\omega t)$ is $V_p\omega$).*
7. **CMRR Analysis:** An Op-Amp has a differential gain $A_d = 100,000$ and a CMRR of 80 dB. If a common-mode noise signal of 10 mV enters the inputs, what voltage will appear at the output due to this noise?
8. **Bandwidth Calculation:** A 741 Op-Amp has a Gain-Bandwidth Product (GBP) of 1 MHz. You design a non-inverting amplifier with a Gain of 50.
   (a) What is the bandwidth (cutoff frequency) of your amplifier?
   (b) If you increase the gain to 500, what happens to the bandwidth?
9. **Input Offset:** An Op-Amp has an Input Offset Voltage $V_{os} = 2$ mV. It is used in a non-inverting configuration with $R_f = 100k\Omega$ and $R_{in} = 1k\Omega$. Calculate the DC error voltage at the output.

10. **Bias Current Error:** An Op-Amp has an input bias current $I_B = 100$ nA. It is used as a voltage follower with a source resistance $R_{source} = 100k\Omega$. Calculate the error voltage introduced at the input due to the bias current flowing through the source resistance.